

NOARLUNGA, Australia

WMO: 948080

Lat: 35.1586S

Lon: 138.5058E

Elev: 55

StdP: 100.67

Time Zone: 9.50 (AUA)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.0	6.9	-4.0	2.7	24.4	-2.1	3.2	23.2	11.8	14.2	10.7	13.1	2.3	60	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.7	36.6	18.2	34.2	17.7	32.0	17.1	20.5	29.3	19.6	28.6	18.8	27.9	4.6	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	22.9	16.7	12.0	22.3	15.5	11.1	21.7	59.1	29.1	56.1	28.7	53.4	27.8	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	10.2	9.0	4.2	41.1	0.6	1.9	3.8	42.5	3.4	43.7	3.0	44.7	2.6	46.2		
(5)			2.8	22.2	0.8	1.0	2.2	23.0	1.7	23.6	1.3	24.1	0.7	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.8	22.3	21.8	20.3	18.0	14.8	12.2	11.7	12.1	13.9	16.1	18.6	20.4	(6)
(7)		DBStd	5.09	4.48	4.22	3.93	3.23	2.36	1.83	1.83	2.30	2.96	3.80	4.42	4.55	(7)
(8)		HDD10.0	12	0	0	0	0	0	3	4	4	1	0	0	0	(8)
(9)		HDD18.3	1079	6	8	18	44	114	184	206	193	138	94	49	25	(9)
(10)		CDD10.0	2506	381	331	320	240	149	69	58	69	119	189	259	322	(10)
(11)		CDD18.3	529	128	105	80	34	4	0	0	1	6	25	58	88	(11)
(12)		CDH23.3	5494	1429	1077	764	236	9	0	0	2	39	273	642	1023	(12)
(13)		CDH26.7	2492	742	501	324	63	1	0	0	0	4	84	270	503	(13)
(14)	Wind	WSAvg	4.5	5.2	5.0	4.6	4.1	3.8	3.7	4.4	4.3	4.7	4.8	4.8	(14)	
(15)	Precipitation	PrecAvg	514	19	19	22	42	67	67	75	65	51	42	26	22	(15)
(16)		PrecMax	748	59	143	87	127	143	134	197	107	122	120	112	53	(16)
(17)		PrecMin	311	0	0	0	2	12	14	20	17	15	4	2	0	(17)
(18)		PrecStd	107	16	27	22	32	33	30	34	26	25	28	19	16	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	38.1	36.4	31.7	24.4	19.9	19.1	22.6	27.0	32.1	35.6	38.4	(19)
(20)			MCWB	19.1	19.6	17.6	16.7	13.8	12.9	11.5	12.4	13.7	15.5	17.3	18.3	(20)
(21)		2%	DB	37.0	34.6	32.7	28.2	22.3	17.5	17.0	19.9	24.3	28.9	32.4	35.3	(21)
(22)			MCWB	18.5	18.6	17.2	15.6	13.5	12.0	10.7	11.5	13.0	14.6	16.4	17.3	(22)
(23)		5%	DB	33.9	32.3	30.4	26.0	20.2	16.1	15.7	17.9	22.1	26.4	29.8	32.1	(23)
(24)			MCWB	17.9	17.8	17.3	14.8	13.2	11.8	10.7	11.0	12.4	13.8	15.8	16.7	(24)
(25)		10%	DB	30.7	29.6	27.7	23.8	18.6	15.2	14.6	16.1	19.8	23.8	27.2	29.1	(25)
(26)			MCWB	17.3	17.4	16.6	14.2	13.0	11.5	10.7	10.6	11.5	13.3	15.4	16.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.7	21.7	20.6	18.3	16.1	14.4	13.5	13.8	15.2	17.4	19.5	20.7	(27)
(28)			MCDWB	28.6	31.9	27.9	23.9	19.8	16.7	16.5	18.3	22.7	26.3	28.0	28.0	(28)
(29)		2%	WB	20.6	20.2	19.5	17.3	15.2	13.4	12.6	12.8	14.2	16.0	18.2	19.3	(29)
(30)			MCDWB	30.3	28.8	26.9	23.1	18.8	15.8	14.9	16.8	20.6	24.6	27.3	28.3	(30)
(31)		5%	WB	19.6	19.3	18.6	16.5	14.5	12.7	12.0	12.1	13.4	15.0	17.4	18.2	(31)
(32)			MCDWB	29.9	28.1	26.6	22.4	18.0	15.2	14.1	15.8	19.2	22.7	25.9	28.0	(32)
(33)		10%	WB	18.5	18.4	17.5	15.6	13.8	12.1	11.5	11.5	12.6	14.2	16.3	17.2	(33)
(34)			MCDWB	28.6	27.7	26.0	21.8	17.4	14.6	13.7	14.9	17.9	21.3	25.2	26.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.7	9.9	9.1	7.8	6.0	5.4	5.1	6.2	7.5	9.2	10.0	10.2	(35)
(36)			MCDBR	15.2	13.4	12.6	10.8	7.5	6.2	5.9	8.7	11.6	14.1	14.2	15.1	(36)
(37)			MCWBR	5.2	5.0	4.7	4.6	4.0	3.9	3.6	4.3	4.8	5.3	5.1	5.5	(37)
(38)		5% WB	MCDWB	12.3	11.4	10.6	8.3	5.8	5.1	4.7	6.7	9.3	11.7	12.3	12.1	(38)
(39)			MCWBR	5.6	5.0	4.8	4.6	3.8	3.7	3.5	4.0	4.6	5.0	5.2	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.342	0.342	0.323	0.317	0.308	0.300	0.299	0.310	0.332	0.343	0.345	0.338	(40)	
(41)		taud	2.549	2.565	2.612	2.613	2.618	2.620	2.616	2.569	2.502	2.477	2.501	2.546	(41)	
(42)		Ebn at Noon	994	973	955	904	858	837	859	897	929	960	984	1002	(42)	
(43)		Edh at Noon	108	102	91	80	70	66	69	82	100	111	113	109	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.71	6.72	5.46	3.95	2.70	2.22	2.41	3.33	4.58	5.95	7.05	7.58	(44)	
(45)		RadStd	0.39	0.33	0.24	0.22	0.11	0.14	0.11	0.20	0.35	0.36	0.31	0.30	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-98	+161	+70

Nomenclature: See separate page